



Sperms move along

To reduce infertility treatment cost, Germans came up with this motorised bot to make sperms move faster.
<http://dgit.in/SpiralBot>



Phone backdoor

New Yorker's privacy may be infringed by the new proposed bill and pave way for other govts. to consider.
<http://dgit.in/PhoneDoor>

compensation to the right-holders. Sounds mighty similar to how music streaming works today, doesn't it?

Jukeboxes and radio continued to be the prime instances of "streaming" music for the longest time, until the advent of the Internet. Towards the end of the millennium, Shawn Fanning started a peer-to-peer file sharing service that shot to massive popularity overnight. Napster, the forefather of music streaming services as we know it today, garnered nearly 80 million registered users in a span of two years. Users could listen to, and download digital files of songs, recording and other audio files which were otherwise difficult to obtain. Napster was an instant hit among the youth, with pre-release singles and entire albums attracting the music-loving youth by the millions.

However, Napster soon ran into copyright infringement lawsuits, which had the world divided on opinion. The service was forced to shut down within three years but, Napster's glory days showed the potential that online sharing and streaming of music had, and laid the foundation for a number of services to follow.

Alongside laying the foundation for music streaming services, Napster's lawsuit involvements also showed the world a possibility of presenting music tracks, albums and channels, for a fee. In 2001, Apple entered the portable music player front with the then-revolutionary iPod. Two years later, the iTunes Music Store was introduced, and with it, the realm of paying for on-demand music opened up. In January 2000, Savage Beast Technologies' Music Genome Project devised multiple characteristics of a track, to help classify and organise songs and tracks by various criteria, like genre, gender of vocalist and level of distortion on guitar, among others. Music Genome Project would go on to become Pandora Media in 2004, and laid the foundation for curating playlists based on listening preference.

By this time, the floodgates had opened for music streaming services to take over. Listen.com (now Rhapsody) premiered in December 2001, while most other services came along towards late-2000s. In the

meantime, Internet Radio services took precedence. November 1994 saw a Rolling Stones concert streamed live, which Mick Jagger opened as saying, "I want to say a special welcome to everyone that's, uh, climbed into the Internet tonight." NetRadio.com became the world's first Internet-only radio service, carving out a niche for itself with massive popularity, which many others followed. Last.fm, Virgin Radio and Broadcast.com were other pioneering Internet radios, along with iTunes Radio. In a bid to take the Internet Radio experience further, Last.fm offered a feature called 'scrobbling', system that tracks what you listen to across different applications like iTunes, Winamp, etc. This was one of the first methods of

personalisation based on listening preferences.

Switching back to music streaming services as we know them today, by the end of 2010, the music-frenzy world saw the likes of Deezer,

Slacker, Saavn, Spotify, Guvera and Rdio enter the music streaming field. Over time, and amid many controversies between the music streaming companies and artists over inadequate royalties, advertisement-supported free music streaming came

to being. Groovespark was a major peg in music licensing controversies. The service used P2P file sharing model, paying users to upload tracks, which were further made available for download, for a fee. Its business model

was approved by a few smaller labels, but lawsuits filed against Groovespark by Universal Music Group, EMI Music Publishing, etc held the service liable for damages worth upto \$736 million if found willfully infringing, following which they service shut down in May 2015.

In spite of these sporadic tussles, users at large could by now listen to tracks, albums and radio stations of their choice, and the songs were only interrupted by advertisements either on screen, or

between songs. This mode of operation allowed the music streaming companies to keep providing free music to worldwide listeners, while generating revenue out of advertisements to fulfill artists' royalties. Keeping the royalty and advertisement

model aside, this is very similar to what Napster had done, almost a decade back – provide free music that can also be downloaded for offline listening.

As of now, two crucial entrants have entered the music streaming industry. While Apple Music has brought with itself a large library of existing artist contracts that it had with iTunes, Tidal has focused primarily on quality. Tidal does not provide free streaming, and instead provides Hi-Fi lossless audio to paying customers. Music streaming services now are introducing multiple-user subscription offers to attract more users, and typically have millions of songs from thousands of artists across multiple genres.

Music streaming services have come a long way – from the very first Telharmonium to the digital music services of today. While we have ventured far away from its origins, the principles remain similar – latest songs from new and established artists alike, diverse playlists and fights over royalties and artist copyrights. In the present day, the focus has diverted to opti-

mising services according to network speeds. While services like Saavn stream low bitrate audio to cater to slower connection speeds, Tidal focuses on high quality audio which consume more data, but provide higher quality of

audio for audiophiles who may run out of storage on portable devices, with the large size of each file.

What began as a novelty in the late 1800s, is a regular product in use today. Music albums premiere over the Internet, concerts are ever so closer since the advent of Internet radio services. What next, then? With virtual reality edging closer towards everyday technology, maybe we'll be streaming concerts from our homes, and yet, be present at venues. 

Radio, only more intelligent



Metallica never approved of Napster spreading their music for free



All that's good always came for a price